

State-of-the-Spine — Keeper audit-side v0.1 (Thursday 2026-05-28 consolidation)

Keeper

2026-05-28 Thursday EDT

State-of-the-Spine — Keeper audit-side

Casey called consolidate (Thursday 2026-05-28). This is the audit/verification layer of the consolidation: the honest tier disposition of the day's convergence, so the papers lead with verified content and label everything else correctly.

1. The two validation axes (the organizing principle of the spine)

A forward claim must pass the axis relevant to *its type*. The two axes are orthogonal (Grace's catch, paired with Cal #31 + Keeper invariant-anchor):

Axis	Test	Applies to	Validates because
Scheme-invariance	dimensionless $\frac{?}{?}$ frame-independent $\frac{?}{?}$ anchored to a geometric/topological invariant	physical observables (masses, angles, ratios)	invariants are frame-independent by nature (Casey's invariant-anchor principle)
Coincidence- denominator	uniquely reachable from the primitives (low # of {rank,N_c}-expressions hitting the target)	arithmetic relations among the integers (the integer web)	a low denominator means it is not back-fit (Elie Toy 3578 / Grace metric; Cal #31 positive partner)

Rule for the papers: every forward claim names which axis validates it. A claim passing neither is a lead, not forward content.

2. Verified forward spine (paper-ready, with tiers)

Content	Tier	Validating axis / basis
Serre structure constants $[2]_2 = N_c$, $[3]_4 = N_c \cdot g(U_q^+(B_2) \text{ at } q=2)$	RIGOROUS	forward-consequence of B_2 Cartan matrix + $q=2$ (Elie + Lyra)
Mixing angles: Cabibbo 9/40; PMNS $\sin^2\theta \in \{42,75,3\}/137$ (sum 120/137); Weinberg $\sin^2\theta_W = \text{rank}/N_c^2 = 2/9$	FRAMEWORK-PLUS	scheme-invariance (dimensionless); $/N_{\text{max}}$ unified structure; Weinberg forced by rank=2

Content	Tier	Validating axis / basis
$m_W/m_Z = \sqrt{g/N_c}$ (0.064%, pole-mass)	FRAMEWORK-PLUS	scheme-invariance (pole-mass ratio, physically privileged)
3 generations = $h(B_2)-1$; 3 colors = $h^v(B_2) = N_c$	FRAMEWORK (count matched, forcing multi-week)	root-system invariants (Elie Toy 3571 verified)
$\rho(D_{IV^5}) = (n_C, N_c)/\text{rank} = (5/2, 3/2)$; ρ_1 bulk (Bergman genus), ρ_2 Shilov (Wallach)	FRAMEWORK-PLUS	canonical spectral invariant (Elie Toy 3583); pins rank/ N_c/n_C , splits bulk/Shilov
Confinement: isolated fractional-charge colored quarks have no Shilov boundary value $\rightarrow$ only color-singlet composites observable	FRAMEWORK-PLUS (Cal “strongest content”)	Hardy-space bulk $\rightarrow$ Shilov projection (Lyra/A3)
Kernel singularity exponent = $5/2 = n_C/\text{rank}$ (Hua genus)	SETTLED	Elie Toy 3582 convention-free $\nu=5$ ($\sim 200\times$ margin; validated on disk $\nu=1$, D_{IV^2} $\nu=2$)
$c_{FK} = 225/\pi^{(9/2)}$ (FK normalized-measure constant)	RATIFIED (T2442), now DERIVED-physical	see Sec 4 — forced by Born-rule invariance
Route A: BST’s 5 integers = standard invariants of D_{IV^5} (rank, $N_c=h^v$, n_C =Hua genus, C_2 =FK genus, g =embedding/signature, $N_{\max}$ =def)	FRAMEWORK-PLUS	“choose D_{IV^5} , zero inputs”; Strong-Uniqueness-supporting

3. Standing conventions adopted today (prevent recurring rechecks)

- Three-genus convention (Grace INV-5262; Lyra A1 §1 + Vol 16 §1.5 + series-wide note): Hua genus = $n_C = 5$; FK genus = $C_2 = 6$; embedding/signature dimension = $g = 7$. Standing rule: *intrinsic quantities (kernel exponent, genus) are never $g=7$; $g=7$ is the embedding signature only.*
- α -disambiguation (Lyra, proactive): three distinct “ α ” — Jack $\alpha_{\text{Jack}} = 2/N_c = 2/3$; fine-structure $\alpha = 1/137$; Macdonald parameter t . Always specify which.
- Both are “specify-which” guards of the same shape as the genus convention — they convert a recurring recheck class into a one-line discipline.

4. c_{FK} / measure-choice — now a DERIVED theorem (not a convention)

The day’s deepest closure. The substrate Hilbert space measure is forced, not chosen: - T754 derives the Born rule as the unique automorphism-invariant probability measure on D_{IV^5} (Gleason-type). - On a bounded symmetric domain, automorphisms have nontrivial Jacobians ∇ Lebesgue is not automorphism-invariant; the unique invariant measure is the Bergman/FK measure. - ∇ physical Hilbert space = $L^2(D_{IV^5}, \text{FK measure})$, forced by Born-rule invariance. $c_{FK} = 225/\pi^{(9/2)}$ is the physical constant (derived); Elie’s $1920/\pi^5$ is the ambient-Lebesgue value (labeled, non-physical).

This is a Route-A / Strong-Uniqueness strengthener: the measure is compelled, not an input.

5. Recheck ledger (the day’s discipline-as-generator outcomes)

All favorable; one root cause ($g=7$ mislabeled as a genus/Bergman quantity); propagated into ≥ 5 places, now all dispositioned:

Item	Outcome	Disposition
c_FK volume constant (T2442)	clean	FK-measure constant, derived-physical (Sec 4); STANDS, relabel “volume”→“FK normalized-measure constant”
all-5-from-B ₂ “g = Bergman exponent”	corrected	g = signature total p+q (forward); n_C = Hua genus (forward); Route A intact
kernel exponent g/rank=7/2	corrected	= n_C/rank = 5/2 (Elie $\nu=5$)
K67/T2401 “Bergman exponent 7/2”	NOT load-bearing	Born rule rests on invariance (T754); outputs $\propto 1/N_{\max}^2, \alpha^2$; STANDS, relabel 7/2→5/2
T2440 “g/rank=7/2”	relabel pending	same disposition; route to Grace 7/2-sweep (INV-5264 extension)

Cross-CI discipline fired in every direction including back at its authors: Keeper (FK-genus naming + premature c_FK closure), Grace (axis-automation accuracy + premature STANDS), Lyra (g=p+2 candidate-not-stated), Elie (Toy 3559 coefficient). Each catch left the framework cleaner — the Calibration #30 pattern (now STANDING).

6. Methodology / calibration advances today

- Calibration #30 → STANDING (honest-negative-strengthens; 27th layer).
- Calibration #31 → STANDING (substrate-finding-as-standard-math-specialization; 28th layer) — with Keeper’s anti-laundersing guard: a specialization counts only if non-generic (low coincidence-denominator) + load-bearing. Matched positive partner to the coincidence-denominator negative test.
- Show-all-threads-then-weave (Casey calibration): the tier-honest brake controls the *claim*, not the *investigation* — keep leads live, weave after.
- Two-axis forward-vs-fitted framework (Grace + Cal #31 + invariant-anchor): Sec 1.

7. Open items + routing (nothing else paper-blocking on the closed threads)

Item	Owner	Status
PMNS form-typing (/N _{max} set vs T1935 4/13,6/11)	Cal	/N _{max} canonical recommended; T1935 alternate; Cal types
Generation-count <i>forcing</i> (Coxeter h(B ₂)=4 matched→forced; cyclotomic↔Coxeter mechanism)	Lyra	multi-week; B1 carries it as open
K67/T2401 + T2440 7/2→5/2 relabel	Grace sweep + Keeper confirm	relabel-only expected; confirm per item
Quark mass ratios (cross-tier)	leads	IDENTIFIED; substrate-privileged-scheme hypothesis live (Lyra #8) — keep investigating per show-all-threads back-fit per coincidence-denominator; not in forward claims
all-5-from-B ₂ Route-B relations (n_C=N _c ² −rank ² , C ₂ =N _c ·rank)	parked leads	multi-week forward thread
Bulk-Shilov unification at Macdonald level (single Macdonald-Koornwinder specializing both ways)	Lyra/Elie	

8. What is NOT yet forward (honest leads — keep OUT of forward claims, keep live as investigations)

- Cross-tier quark mass ratios (scheme-dependent; $m_t/m_c \approx N_{\text{max}}$ etc. — leads, not derivations).
- Macdonald-coefficient $\leftrightarrow$ mass-ratio link (IDENTIFIED; needs mechanism + scheme-invariant relation).
- Generation count as *forced* (matched only).
- Route-B algebraic primary relations (back-fit).
- The fermion mass-web / winding-step program (Casey seed; criterion strong, web is a program).

9. Keeper sign-off

The day's verified spine is honest and paper-ready: lead with the scheme-invariant mixing-angle block, the rigorous Serre constants, the ρ -vector / Route-A reduction, the confinement mechanism, and the derived FK-measure c_{FK} . Label the leads as leads. Every forward claim names its validating axis. The three rechecks all resolved favorably and *strengthened* the framework (measure now derived; ρ -vector pins three primaries; conventions prevent recurrence).

No premature RATIFIED. Tier discipline held throughout. This audit-side is ready to fold into Lyra's narrative spine and to seed the paper methodology sections.

— Keeper, 2026-05-28 Thursday (state-of-the-spine, audit-side v0.1)